

Ex. No. 3 — Designing a Quarter-Wave Transformer by Varying a) Frequency b) Input Impedance c) Characteristic Impedance

Test Case 1 — Varying Frequency

Design a quarter-wave transformer for an input impedance $Z = 50 \Omega$ and a load impedance $Z = 100 \Omega$.

in

L

Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

Scilab Program

```
clear; clc; Zr =  
100; Zs = 50;  
f = 100*10^6; // set frequency per row (Hz) Ro  
= sqrt(Zs*Zr); L = 300/(f*10^-6); printf('length of  
lambda/4 transformer = %f meters\n', L);  
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```

The screenshot displays the Scilab 2026.1.0 interface. On the left, the 'Console' window shows the output of the program: 'length of lambda/4 transformer = 0.600000 meters' and 'characteristic impedance of quarter wave line = 70.710600'. On the right, the 'Editor' window shows the source code for 'exp 3.sci'. The code defines variables for input impedance (Zr=100), load impedance (Zs=50), and frequency (f=100 MHz). It calculates the characteristic impedance (Ro) and the length of the quarter-wave transformer (L) in meters. The results are printed using the 'printf' function.

```
1 clear;clc;  
2 Zr=100;s=9;d=0.1;  
3 Zs=50;  
4 Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;  
5 Zof=276*log10(s/r);  
6 Zoq=sqrt(Zr*Zof);  
7 L=300/(f*(10^-6));  
8 printf("length of lambda/4 transformer = %f meters\n",L);  
9 printf("characteristic impedance of quarter wave line = %f.",fix(Ro*10000)/10000);  
10
```

Sample Console Output

```
length of lambda/4 transformer = 0.600000 meters
characteristic impedance of quarter wave line = 70.710600
-->
```

Characteristic Impedance of the Quarter-Wave Line

Z _L (ohms)	Z _{in} (ohms)	Simulation R _{in} (ohms)	Theoretical R _{in} (ohms)
50	100	70.7107	70.7107

Length of λ/4 Transformer vs Frequency (Z = 100 Ω, Z = 50 Ω) L_{in}

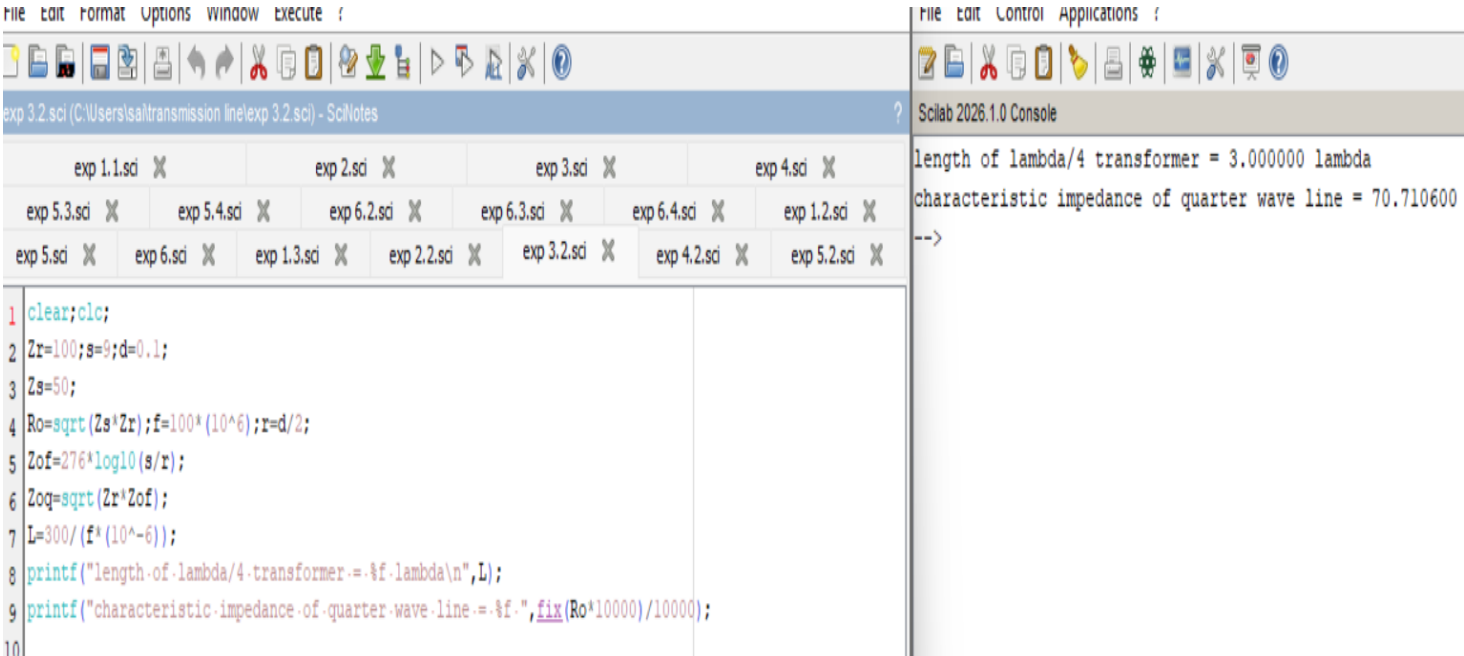
Frequency (MHz)	Simulation Length (m)	Theoretical Length (m)
100	3.0000	3.0000
200	1.5000	1.5000
300	1.0000	1.0000
500	0.6000	0.6000

Test Case 2 — Varying Input Impedance

For a fixed operating frequency of 2 GHz and a load impedance $Z = 100\ \Omega$, design a quarter-wave transformer L and evaluate its performance as the input impedance Z varies through $\{25\ \Omega, 50\ \Omega, 75\ \Omega, 100\ \Omega\}$. Validate in that the transformer's characteristic impedance ($Z_o = R_o = \sqrt{(Z \cdot Z_L)}$) changes with Z . transformer in

Scilab Program

```
clear;clc; Zr = 100;
f = 2*10^9;
Zs = 25; // set Zin per row
Ro = sqrt(Zs*Zr); L = 300/(f*10^-6); printf('length of
lambda/4 transformer = %f lambda\n', L);
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```



Sample Console Output

```
length of lambda/4 transformer = 3.000000 lambda
characteristic impedance of quarter wave line = 70.710600
-->
```

R_{in} vs Z_L (Z = 100 Ω, f = 2 GHz) in

Z _L (ohms)	Simulation R _{in} (ohms)	Theoretical R _{in} (ohms)
25	50.0000	50.0000
50	70.7107	70.7107
75	86.6025	86.6025
100	100.0000	100.0000

Result A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (Z_L) of 100 Ω, and its performance is evaluated for various input impedance values. The variation of the transformer's characteristic impedance with Z_L is also validated.